Supplemental Materials

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This document provides supplemental materials for the manuscript entitled "Diffeomorphic Coverage Control in Non-Convex Annulus Region with Conformal Mapping " by Xun Feng, Chao Zhai, Chuan-Ke Zhang, and Yong He, and it is composed of two parts. The first part presents some theoretical results, and it includes some key definitions and theoretical proof of key lemmas, corollaries and theorem. The second part provides simulation results on the original environment and relevant discussions, which includes the analysis of computational cost, quantitative metrics, comparisons with baseline approaches, scenarios of multiple obstacles, scalability of the algorithm and the effect of agent failure.

1 Theoretical Results

1.1 Key definitions

First of all, the definitions of the article are introduced below.

Definition 1.1. *(Conformal Mapping [Gu, 2007]). Let $f:\mathbb{C}\to\mathbb{C}$ be a mapping with $f(z)=f(x,y)=u(x,y)+iv(x,y)$, where u, v are real-valued functions. f is said to be a conformal map if it is differentiable everywhere in $\mathbb{C}$, for any $z\in\mathbb{C}$, $f'(z)\neq 0$, and f preserves angle. In other words, f satisfies the Cauchy–Riemann equations*

$$\frac{\partial u}{\partial x}=\frac{\partial v}{\partial y},\quad \frac{\partial v}{\partial x}=-\frac{\partial u}{\partial y}.$$

Definition 1.2. *(Quasi-Conformal Mapping [Gu, 2007]). A mapping $f:\mathbb{C}\to\mathbb{C}$ is said to be a quasi-conformal mapping if it satisfies the Beltrami equation*

$$\frac{\partial f}{\partial \bar{z}}=\mu_f(z)\frac{\partial f}{\partial z}$$

for some complex-valued function μ_f with $\|\mu_f\|_\infty < 1$, where the complex derivatives are given by

$$\begin{cases} \dfrac{\partial f}{\partial z} = \dfrac{1}{2}\left(\dfrac{\partial f}{\partial x} - i\dfrac{\partial f}{\partial y}\right) \\ \dfrac{\partial f}{\partial \bar{z}} = \dfrac{1}{2}\left(\dfrac{\partial f}{\partial x} + i\dfrac{\partial f}{\partial y}\right) \end{cases}$$

where μ_f is called the Beltrami coefficient of f. Essentially, quasi-conformal mapping is used to reduce the angle distortion in the design process of conformal mapping.

Definition 1.3. *(Orientable Altas [Petersen, 2006]). Suppose A is an Altas in Ξ, we call A an orientable altas if any two coordinate charts in A are oriented-compatible.*

Definition 1.4. *(Orientable Manifold [Petersen, 2006]). Ξ is an orientable manifold if there exists an oriented-compatible altas in Ξ.*

Definition 1.5. *(Cut Locus [Wolter, 1985]). Let $p \in \Omega$ and $q \in (\Omega - \partial\Omega)$, we call q is a pica relative to p if there are more than one minimal paths that can connect p and q. The closure of the set of all pica relative to p is called the cut locus of p, which is denoted as C_p.*

Although the cut locus is a troublesome set that contributes to the non-uniqueness of the optimal path, we can prove that it is a Lebesgue null set in Lemma3.3.

Definition 1.6. *(Geodesically Convex [Petersen, 2006]). Suppose (Ω, η) is a connected Riemannian manifold with Riemannian metric η, and there is a unique minimum geodesic connecting p and q, which is entirely within this region. We call Ω a geodesically convex region.*

There exists a unique minimum geodesic connecting any two points in a geodesically convex region. Actually, the key technical challenge of this work is to establish the geodesically convex nature of the coverage region, which is essential to analyze the existence of the optimal path.

1.2 Conformal mapping

This subsection provides additional details regarding the calculation process and proof related to conformal mapping. The Uniformization Lemma significantly simplifies the study of simply connected Riemannian surfaces by reducing it to the study of the disk, plane, or sphere.

Lemma 1.1 (Uniformization Lemma [Gu, 2007])**.** *Every simply connected Riemannian surface is conformally equivalent to the unit disk, complex plane, or Riemannian sphere.*

Then, we provide the computational process of the harmonic mapping H_i. For the i-th agent in E_i, compute a disk harmonic mapping $H_i : E_i \to \mathbb{D}$ by solving the following Laplace equation

$$\Delta H_i = 0, \quad H_i(\partial E_i) = \partial \mathbb{D}, \tag{1}$$

where Δ is the Laplace–Beltrami operator on E_i, and it can be easily discretized using the cotangent Laplacian [Meng *et al.*, 2016]. This discretization transforms the Laplace equation $\Delta H_i = 0$ into a sparse, symmetric positive definite linear system, which allows to readily obtain the solution. The Beltrami coefficient $\mu(H_i) = \lambda_i + j\gamma_i$ measures the conformality distortion of H_i. In fact, a smaller norm of the Beltrami coefficient implies a tinier conformality distortion. With the Dirichlet boundary conditions [Lui *et al.*, 2013], the problem of solving the disk harmonic mapping can be converted to solve the partial differential equations

$$\nabla \cdot \left(A \begin{pmatrix} (u_i)_x \\ (u_i)_y \end{pmatrix} \right) = 0, \ \nabla \cdot \left(A \begin{pmatrix} (v_i)_x \\ (v_i)_y \end{pmatrix} \right) = 0$$

with

$$A = \begin{pmatrix} \alpha_1 & \alpha_2 \\ \alpha_2 & \alpha_3 \end{pmatrix}, \ \alpha_1 = \frac{(\lambda_i - 1)^2 + (\gamma_i)^2}{1 - (\lambda_i)^2 - (\gamma_i)^2},$$

$$\alpha_2 = -\frac{2\gamma_i}{1 - (\lambda_i)^2 - (\gamma_i)^2}, \quad \alpha_3 = \frac{(\lambda_i + 1)^2 + (\gamma_i)^2}{1 - (\lambda_i)^2 - (\gamma_i)^2}.$$

By solving the above equations, we obtain the x-coordinate and y-coordinate functions of H_i, which are used to obtain the coordinates of the point cloud set v^i in the unit disk $\mathbb{D}$.

Next, we provide the computational process of the quasi-conformal mapping σ_i. The Beltrami coefficient of σ_i satisfies

$$\mu_{\sigma_i} = \mu_{H_i^{-1}}. \tag{2}$$

In numerical computation, the Linear Beltrami Solver (LBS) method is employed to compute the quasi-conformal map σ_i [Lui *et al.*, 2013]. The x-coordinate and y-coordinate of i-th agent are updated by

$$\begin{cases} x_+ = L_i x_- \\ y_+ = y_- \end{cases}$$

Here, x_+ and y_+ are used to obtain the norm of the Beltrami coefficient of $\vartheta_i = \sigma_i \circ H_i$, implying the length L_i affects the norm of Beltrami coefficient. To reduce the conformal distortion, we need to minimize the norm of the Beltrami coefficient. Thus, the optimal index is proposed as follows

$$L_* = \arg\min \|\mu_\vartheta\|, \tag{3}$$

with $\mu_\vartheta = (\mu_{\vartheta_1}, \cdots \mu_{\vartheta_N})$. The optimum length L_* minimizes the norm of the Beltrami coefficient μ_ϑ, indicating the minimal conformal distortion. Under the topology of uniform convergence on the region, the function space composed of rectangular conformal maps is compact [Gu, 2007]. Since the function space is compact, the existence of the optimal index (3) is guaranteed.

Remark 1.1. *According to Definition 1.2, because the rectangular conformal mapping ϑ_i satisfies $\|\mu_{\vartheta_i}\|_\infty < 1$, the uniform boundedness of ϑ_i is ensured. Since the mapping ϑ_i is defined on a compact region $\mathbb{D}$, ϑ_i is equicontinuous. According to Arzelà–Ascoli Lemma [Gu, 2007], the function space composed of rectangular conformal maps ϑ_i is relatively compact. Given that the function space is closed, we can prove the compactness of the function space.*

After that, we provide the calculation method of the Jacobian J_f. In the complex plane $\mathbb{C}$, note that $f(z) = f(x,y) = u(x,y) + iv(x,y)$ with u, v are real-valued functions, and the Jacobian J_f of quasi-conformal mapping f is given by

$$J_f = |f_z|^2 \left(1 - \left|\mu_f\right|^2\right), \tag{4}$$

where μ_f is the Beltrami coefficient of f. This concept can be naturally extended to the Riemannian manifold with the aid of local charts. Finally, we provide the proof of Lemma 1.2, which corresponds to Lemma 3.1 on page 4 of the revision scripts.

Lemma 1.2. *The properties of conformal mapping τ.*

1. *τ is an orientation-preserving diffeomorphism.*
2. *The discretization error of $(\Omega - O) \cup \partial O$ with Delaunay triangulation can be arbitrarily small.*
3. *Delaunay triangulation and diffeomorphism τ guarantee the existence of partition phases in (Ω-O)$\cup\partial O$.*
4. *Conformal diffeomorphism τ preserves the partition angle of each subregion.*
5. $(m_i - m_{i-1}) \cdot \left(m_i' - m_{i-1}'\right) > 0$.

Proof. For Claim 1), according to Definition 1.2, the mapping ϖ satisfies $\|\mu_\varpi\|_\infty < 1$. Based on (4), the Jacobian determinant of the quasi-conformal mapping $\det(J_\varpi)$ is positive. Since the Jacobian determinant of conformal mapping is positive, we obtain $\det(J_{\vartheta^*}) > 0$. According to $\det(J_\iota) > 0$ and $\tau = \varpi \circ \iota \circ \vartheta^*$, we have $\det(J_\tau) > 0$, indicating that τ is an orientation-preserving diffeomorphism.

For Claim 2), in the process of mapping design, Delaunay triangulation is used to discretize the region (Ω-O)$\cup\partial O$. When each agent is driven to its optimal location, it happens that the optimal point is inside the Delaunay triangle, which means the point is not the point cloud set and thus leads to the discretization error. Denote the actual solution and the optimal solution as v_a and v_o, respectively. In the 2-dimensional compact manifold (Ω-O)$\cup\partial O$, we set point $v_o = (x_o, y_o)$. Without loss of generality, suppose that the maximum difference between v_a and v_o is $\|x_a - x_o\|_\infty$ with the numerical difference $\|x_a - x_o\|_\infty \leq h < 1$, where h is the meshing size and $p > 1$ is a convergence index. It satisfies that for any $\varepsilon > 0$, there exists $\delta = (\varepsilon/\sqrt{n})^{\frac{1}{p}}$, such that $\|v_a - v_o\|_\infty \leq \sqrt{n}\|v_a - v_o\|_\infty \leq \sqrt{n}\|x_a - x_o\|_\infty < \sqrt{n}\|x_a - x_o\|_\infty^p \leq \sqrt{n}\delta^p = \varepsilon$ when $\|x_a - x_o\|_\infty < \delta$. Then we obtain $\|v_a - v_o\|_\infty < \sqrt{n}h^p$. According to the error convergence analysis in [Eca *et al.*, 2009], the discretization error of $(\Omega - O) \cup \partial O$ with Delaunay triangulation can be arbitrarily small by tuning the mesh size h.

For Claim 3), the triangular mesh structure generated by Delaunay triangulation is preserved under the diffeomorphism τ, indicating that the connectivity of regions remains unchanged. This implies that τ maintains the adjacency of triangles formed by Delaunay triangulation in $(\Omega - O) \cup \partial O$, ensuring they remain adjacent in Ξ. Note that (Ω-O)$\cup\partial O$ is topologically equivalent to an annular region and diffeomorphism τ preserves the topology of the surface. Consequently, Delaunay triangulation of the boundary is maintained on the boundary after the mapping τ, which guarantees that boundary points in $(\Omega - O) \cup \partial O$ are mapped to the boundary of Ξ. More explicitly, the mapping τ satisfies $\tau : \mathrm{int}\left((\Omega - O) \cup \partial O\right) \to \mathrm{int}\Xi$ and $\tau : \partial\left((\Omega - O) \cup \partial O\right) \to \partial\Xi$, where $\mathrm{int}\Xi$ denotes the inner region of Ξ, and $\partial\Xi$ represents the outer region of Ξ. The partition bars are continuous curves that connect the inner and outer boundaries in Ξ. Due to the continuity of τ^{-1} and the correspondence relationship of boundaries, the existence of partition bars is guaranteed in $(\Omega - O) \cup \partial O$.

For Claim 4), we analyze the impact of τ on the partition bars. According to Claim 3), the Delaunay triangulation guarantees the continuity of the length change of tangent vectors between adjacent triangles, and the connectivity of regions remains unchanged. Since τ is an orientation-preserving diffeomorphism based on Claim 1), it does not alter the local relative angle sequence (i.e. $\psi_1(0) < \psi_2(0) < \cdots < \psi_N(0)$ if $\psi_1'(0) < \psi_2'(0) < \cdots < \psi_N'(0)$). Moreover, it preserves the local distribution of Delaunay triangles. All of Delaunay triangles in chart (U_α, ϕ_α) will be mapped to $(\tau(U_\alpha), \phi_\alpha')$ with $U_\alpha \subset (\Omega - O) \cup \partial O$ and $\tau(U_\alpha) \subset \Xi$. Thus, the relative positions of partition bars can be maintained in $(\Omega - O) \cup \partial O$ and Ξ. Because the conformal diffeomorphism τ preserves the angle of the Delaunay triangle and the one-to-one correspondence relationship of the Delaunay triangle in boundary, the partition phase satisfies $\psi_i = \psi_i'$.

For Claim 5), according to Claim 4), we obtain $\psi_i = \psi_i'$. Since the sequential number of partition phases in $(\Omega - O) \cup \partial O$ and in Ξ are accurately consistent, we have $\dot{\psi}_i = \dot{\psi}_i'$. Since $k_{\psi'}$ and k_ψ are positive constants, it follows from $\dot{\psi}_i = k_\psi(m_i - m_{i-1})$ and $\dot{\psi}_i' = k_{\psi'}\left(m_i' - m_{i-1}'\right)$ that $(m_i - m_{i-1}) \cdot \left(m_i' - m_{i-1}'\right) > 0$. This completes the proof.

Remark 1.2. *Ξ and (Ω-O)$\cup\partial O$ are topologically equivalent, and the orientation is a topological invariant. Suppose A is an orientable altas in Ξ. Since τ is an orientation-preserving diffeomorphism, there must exist an orientable altas in (Ω-O)$\cup\partial O$.*

Remark 1.3. *Claim 4) in Lemma 3.1 guarantees the boundary correspondence between $(\Omega - O)\cup\partial O$ and Ξ. This implies that the preimage of the minimal path within Ξ is entirely located within $(\Omega - O)\cup\partial O$.*

1.3 Geometric properties

Firstly, we provide the proof of Lemma 1.3, which corresponds to Lemma 3.2 on page 5 of the revision scripts.

Lemma 1.3. *Suppose U is an open set in $\mathbb{R}^2$, $f : U \to \mathbb{R}^2$, $f \in C^1(U)$. y_0 is the regular value of f. If $y_0 \in f(U)$ and $U \supset \Xi = f^{-1}(y_0)$, Ξ is an orientable manifold.*

Proof. Note that U is a region in $\mathbb{R}^2$ with $f \in C^1(U)$. According to Regular Value Theorem [Petersen, 2006], if $y_0 \in f(U)$ and $U \supset \Xi = f^{-1}(y_0)$, Ξ must be a manifold. Note that $f' = (\partial f/\partial x^1, \partial f/\partial x^2) = \nabla f$, $\mathrm{rank}(f') \equiv 1$ in Ξ because y_0 is the regular value of f, which means that $\nabla f \neq 0$ is a normal vector of Ξ. $\vec{n} = \nabla f/\|\nabla f\|$ forms a continuous unit normal vector field in Ξ, and the continuity is guaranteed by $f \in C^1(U)$. The set (x^1, x^2) is the coordinate representation in U. Suppose A is an atlas in Ξ, (U_α, ϕ_α) and (U_β, ϕ_β) are coordinate charts in A and $U_\alpha \cap U_\beta \neq \emptyset$. $\phi_\alpha : I^1 \to U_\alpha$, $t^1 \mapsto (x^1(t^1), x^2(t^1))$, $\phi_\beta : I^1 \to U_\beta$, $s^1 \mapsto (x^1(s^1), x^2(s^1))$, where I^1 is the 1-dimensional unit cube, t^1 and s^1 are the coordinate representation in I^1. The tangent vector of the coordinate chart is represented as $\partial/\partial t^1 = (\partial x^1/\partial t^1, \partial x^2/\partial t^1)$. We call (U_α, ϕ_α) a proper chart if $(\vec{n}, \partial/\partial t^1)$ is oriented-compatible with $\mathbb{R}^2$ standard basis, otherwise we call it an improper chart. For any improper chart, replace t^1 with $-(t^1)$, then it becomes a proper chart. Suppose all coordinate charts in A are proper, then it satisfies that $\det(\vec{n}, \partial/\partial t^1) > 0$ and $\det(\vec{n}, \partial/\partial s^1) > 0$ with $\partial x^i/\partial s^1 = (\partial t^1/\partial s^1)(\partial x^i/\partial t^1)$, $i = 1, 2$, where $\partial t^1/\partial s^1 = \phi'_{\alpha\beta}$, $\phi_{\alpha\beta} = \phi_\alpha^{-1} \circ \phi_\beta$. Thus, we obtain

$$\left(\vec{n}, \partial/\partial t^1\right) = \left(\vec{n}, \partial/\partial s^1\right) \begin{pmatrix} 1 & 0 \\ 0 & \partial t^1/\partial s^1 \end{pmatrix}.$$

By taking the determinants on both sides of the equation, we have $\det(\partial t^1/\partial s^1) > 0$, and it is equivalent to $\det(\phi'_{\alpha\beta}) > 0$. This implies that A is an oriented-compatible altas. According to Definition 1.4, Ξ is an orientable manifold. $\square$

Then, we provide the proof of Lemma 1.4, which corresponds to Lemma 3.3 on page 5 of the revision scripts.

Lemma 1.4. *Subregion E_i is a geodesically convex region.*

Proof. (E_i, d_l) is a smooth manifold with a boundary. Claim 3) in Lemma 3.1 indicates the existence of the partition phase ψ_i in the region. Moreover, it is a complete metric space with the length metric. According to the Hopf-Rinow Theorem [Petersen, 2006], there exists a minimal path (though not necessarily unique) connecting any pair of points. Consider a point $p \in E_i$ and another point $q \in E_i - \partial E_i$. To ensure the uniqueness of the minimal path between p and q, the cut locus C_p should be a Lebesgue null set. For any point $p \in E_i$, since E_i can be considered as a subset of a smooth, complete manifold of the same dimension, according to Theorem 6.2 in [Wolter, 1985], the cut locus C_p is a Lebesgue null set in E_i. Consequently, there exists only one minimal path connecting p and q, which is entirely within this subregion, indicating that the subregion E_i is a geodesically convex region. □

Corollary 1.1. *$(\Omega - O) \cup \partial O$ is an orientable manifold, and the boundaries of $(\Omega\text{-}O)\cup\partial O$ are also orientable.*

Proof. Let $\{(U_\alpha, \phi_\alpha)\}$ be an orientable altas in Ξ. For any point $x \in \Xi$, there exits a diffeomorphism $\tau^{-1} : \Xi \to (\Omega - O) \cup \partial O$. Let $y = \tau^{-1}(x)$, there exists a chart (U_α, ϕ_α) in Ξ such that $x \in U_\alpha$, $y \in \tau^{-1}(U_\alpha)$. Since τ is an orientation-preserving diffeomorphism, $\{(\tau^{-1}(U_\alpha), \phi_\alpha \circ \tau)\}$ is an orientable altas in $(\Omega\text{-}O)\cup\partial O$. According to Definition 1.4, $(\Omega\text{-}O)\cup\partial O$ is an orientable manifold. Since $\mathbb{R}^2 \supseteq \Xi$ is a 2-dimensional orientable manifold with boundary, $\partial\Xi$ is a 1-dimensional orientable manifold without boundary [Petersen, 2006]. This indicates that the boundaries of Ξ are also orientable. Note that orientation is a topological invariant, which in turn implies that the boundaries of $(\Omega\text{-}O)\cup\partial O$ are orientable. □

Corollary 1.2. *The region $(\Omega - O) \cup \partial O$ is a geodesically convex region.*

Proof. Note that E_i is a geodesically convex region and $E_i \subseteq (\Omega - O) \cup \partial O \subseteq \sum_{i=1}^{N} E_i$. In addition, the partition bar with $\psi_i = E_{i-1} \cap E_i$ is a continuous and smooth set, which means that the Riemannian metric carried by each subregion is continuous at the splicing point, and the geodesic is coincident when the subregions E_i and E_{i-1} are spliced together. Thus, the minimal path passing through different subregions is entirely within the region $(\Omega - O) \cup \partial O$. This implies that the region $(\Omega - O) \cup \partial O$ is also a geodesically convex region. □

Likewise, the mapped region Ξ and subregion E_i' are also geodesically convex when we use $d_l'(p,q)$ to measure the distance between p and q. Since

$$\sqrt{\eta_{ij}(\phi(q))\frac{dx^i}{dt}\frac{dx^j}{dt}} = \frac{\eta_{ij}(\phi(q))\frac{dx^i}{dt}\frac{dx^j}{dt}}{\sqrt{\eta_{mn}(\phi(q))\frac{dx^m}{dt}\frac{dx^n}{dt}}},$$

if the position of agent p is not within the chart (U,ϕ), we obtain

$$\frac{\partial}{\partial q^i} d_l^{\tilde{C}}\left(p,\phi\left(q\right)\right) = \frac{\eta_{ij}\left(\phi\left(q\right)\right) z^j_{p\phi(q)}}{\sqrt{\eta_{mn}\left(\phi\left(q\right)\right) z^m_{p\phi(q)} z^n_{p\phi(q)}}}, \tag{5}$$

where $z_{p\phi(q)} = [z^1_{p\phi(q)}, z^2_{p\phi(q)}, \cdots, z^N_{p\phi(q)}]^T$ is the coefficient vector of the tangent vector at q to the shortest path connecting $p \in ((\Omega - O) \cup \partial O) - U$ to $q \in U$. The following remark gives a brief explanation of (5).

Remark 1.4. *Let $Q = \phi(q)$. Consider a minimal path $\gamma_{pq} = \gamma_{pw} \cup \gamma_{wq}$, where $w \in B_q$, B_q is a open ball of q, γ_{pw} is a minimal path from p to w, and γ_{wq} is a minimal geodesic from w and q. For any $Q \in \phi(B_q)$, we have $d_l^{\tilde{D}}\left(p,Q\right) \le d_l\left(p,w\right) + d_l^{\tilde{D}}\left(w,Q\right)$. The equality holds when p,w and $\phi^{-1}(Q)$ lie on the same short path. According to Corollary 1.2 and Theorem 6.2 in [Wolter, 1985], the original region $(\Omega - O) \cup \partial O$ is a geodesically convex region, which implies that the cut locus C_{p_i} is a Lebesgue null set in chart (U,ϕ). This implies that the differentials of the length metric at Q should be the same, which leads to*

$$\frac{\partial}{\partial Q} d^{\tilde{D}}\left(p,Q\right) = \frac{\partial}{\partial Q} d^{\tilde{D}}\left(w,Q\right).$$

According to the derivative analysis in [Pimenta et al.*, 2008], the coefficient vector of the tangent vector at Q is the same, which equals to $z_{pQ} = z_{wQ}$.*

Finally, we provide the computation process of $L(\gamma)$. Let $L(\gamma)$ denote the path length of γ, and it is equal to the integral of the square root of the sum of the products of the coefficients $\eta_{ij}(\gamma(t))$ and the squares of the derivatives of the coordinates x^i with respect to t, over the interval $[p,q]$.

1.4 Details of coverage controller design

The workload on the i-th subregion can be calculated by

$$m_i = \int_{\psi_i}^{\psi_{i+1}} \omega(\theta) d\theta, \quad \omega(\theta) = \int_{l_{in}(\theta)}^{l_{out}(\theta)} \rho(l,\theta) l dl \tag{6}$$

The control input in $(\Omega - O) \cup \partial O$ is designed as

$$u_i = \int_{E_i(\psi) - \left(p_i \cup \psi_i \cup C_{p_i}\right)} \frac{-2k_p d_l\left(q,p_i\right) z^l_{qp_i}}{\sqrt{\eta_{mn}\left(p_i\right) z^m_{qp_i} z^n_{qp_i}}} \rho\left(q\right) dq, \tag{7}$$

First, we provide the detailed proof of Theorem 1.1, which corresponds to Theorem 3.1 on page 6 of the revision scripts.

Theorem 1.1. *Partition dynamic* (17) *equalizes workload on each subregion of the original space $(\Omega - O) \cup \partial O$.*

Proof. Construct the Lyapunov function candidate as $V(\psi) = \frac{1}{2}\sum_{i=1}^{N}(m_i - \bar{m})^2$, where $\bar{m} = \frac{1}{N}\int_0^{2\pi}\omega(\theta)d\theta$ and $V(\psi) \geq 0$. From (6), we understand that $\dot{m}_i = \dot{\psi}_{i+1}\omega(\psi_{i+1}) - \dot{\psi}_i\omega(\psi_i)$. Due to $\psi_{N+1} = \psi_1$, we obtain $\sum_{i=1}^{N}\dot{m}_i = 0$. According to Claim 4) in Lemma 3.1, we have $\psi_i = \psi_i'$. Note that $(\Omega - O) \cup \partial O$ is an orientable manifold, which implies that these charts are oriented-compatible. Then, we can choose an orientation of the region $(\Omega - O) \cup \partial O$. According to Claim 4) in Lemma 3.1, the relative position of partition phases in $(\Omega - O) \cup \partial O$ exactly matches with that in Ξ. Thus, we deduce that $\dot{\psi}_{N+1}' = \dot{\psi}_{N+1}$. Adhering to the cyclic ordering rule, we reinterpret the last term as the first, ensuring that it satisfies $\sum_{i=1}^{N} m_i \dot{\psi}_{i+1}'\omega(\psi_{i+1}) = \sum_{i=1}^{N} m_{i-1}\dot{\psi}_i'\omega(\psi_i)$. Since the conformal map τ is a diffeomorphism and collision avoidance of partition phases in Ξ is guaranteed [Zhai *et al.*, 2023], collision avoidance of partition phases in $(\Omega - O) \cup \partial O$ is guaranteed. Moreover, the time derivative of the Lyapunov function along the system is given by

$$
\begin{aligned}
\frac{dV(\psi)}{dt} &= \sum_{i=1}^{N}(m_i - \bar{m})\dot{m}_i \\
&= \sum_{i=1}^{N} m_i\left(\dot{\psi}_{i+1}\omega(\psi_{i+1}) - \dot{\psi}_i\omega(\psi_i)\right) \\
&= \sum_{i=1}^{N} m_i\left(\dot{\psi}_{i+1}'\omega(\psi_{i+1}) - \dot{\psi}_i'\omega(\psi_i)\right) \\
&= -\sum_{i=1}^{N}(m_i - m_{i-1})\dot{\psi}_i'\omega(\psi_i) \\
&= -K_{\psi'}\sum_{i=1}^{N}\left(m_i' - m_{i-1}'\right)(m_i - m_{i-1})\,\omega(\psi_i).
\end{aligned}
$$

According to Claim 5) in Lemma 3.1, we have $(m'_i - m'_{i-1})(m_i - m_{i-1}) > 0$. Since $K_{\psi'} > 0$ and $\omega(\psi_i) > 0$, it implies that $dV(\psi)/dt < 0$. Therefore, the system is asymptotically stable, and the state of the system will approach the equilibrium point as time goes to infinity, which implies that $m_1 = m_2 = \ldots = m_N$. $\square$

Remark 1.5. *Equation* (7) *elucidates the relationship between the control input and length metric. Then we need to prove that the proposed control input can achieve collision avoidance among agents. This can be accomplished by defining the minimum path that connects the agent position to the optimal centroid using the length metric, rather than the Euclidean metric [Kumar* et al.*, 2014].*

The following remark illustrates that ψ_i is a Lebesgue null set.

Remark 1.6. *In addition, the partition bar with the phase ψ_i is a Lebesgue null set, which can be reduced to the existence of a Lebesgue null set within the unit cube $I^2 \subset \mathbb{R}^2$. Suppose that for any chart U, $\mu : I^2 \to U \subseteq (\Omega - O) \cup \partial O$, $\mu^{-1}(\psi_i \cap U)$ is a Lebesgue null set within I^2. Since μ is a C^1*

diffeomorphism and C^1 diffeomorphism preserves the Lebesgue null measure property, ψ_i is a Lebesgue null set.

1.5 The feasibility of the higher and nonlinear dynamics

For the higher-order dynamics, the dynamics of the agent can be the n-th($n \geq 2$) integrator as follows

$$p_i^{(n)} = u_i, \tag{8}$$

where u_i is the control input and $i \in \{1, \cdots, N\}$, and it can be designed as

$$u_i = -k_p \frac{\partial J}{\partial p_i} \eta^{il}(p_i) - \sum_{k=1}^{n-1} k_{d,k} \left(\frac{D^k p}{Dt^k} \right)_i, \tag{9}$$

where $D^k p/Dt^k$ is a k-th order covariant derivative of p with respect of t, and $k_{d,k}$ is positive constant. The specific computation process of the k-th order covariant derivative $D^k p/Dt^k$ are illustrated in [Petersen, 2006]. According to the Hopf-Rinow Theorem and the Weierstrass Approximation Theorem, the trajectories of agents exist, and those trajectories can be approximated by a polynomial in $(\Omega - O) \cup \partial O$ [Petersen, 2006]. Then, the convergence of the dynamics is guaranteed by Theorem 2 in [Chen *et al.*, 2020]. For the nonlinear dynamics, the dynamic of the agent is described by

$$\dot{p}_i(t) = f(p_i(t), t) + u_i(t), \tag{10}$$

where u_i is the control input, $i \in \{1, \cdots, N\}$. And the function $f(p_i(t), t)$ satisfies the Assumption 2 and 3 in [Ning *et al.*, 2018]. Then the control input u_i is designed as

$$u_i = -k_p \frac{\partial J}{\partial p_i} \eta^{il}(p_i) - k_\beta \left[\frac{\partial J}{\partial p_i} \eta^{il}(p_i) \right]^{1-\frac{a}{b}} - k_\gamma \left[\frac{\partial J}{\partial p_i} \eta^{il}(p_i) \right]^{1+\frac{a}{b}}, \tag{11}$$

where a and b are a positive even integer and a positive odd integer with $a < b$, respectively. And k_p, k_β and k_γ are tunable positive constants. According to Theorem 3 in [Ning *et al.*, 2018], the control input 11 enables the system (10) to converge to the optimal solutions in a fixed time.

1.6 The adaptivity of the dynamical environment

We analyze the adaptivity of the designed dynamic and the controller in the dynamic environment whose density varies slowly. At first, the density function $\rho : (\Omega - O) \cup \partial O \to R^+$ is continuous in the compact region $(\Omega - O) \cup \partial O$. Inspired by [Schwager *et al.*, 2007], we assume that there exists an ideal parameter vector a such that

$$\rho(q) = \kappa(q)^T a. \tag{12}$$

Then, the sensory function approximation for agent i is given by

$$\hat{\rho}_i(q,t) = \kappa(q)^T \hat{a}_i(t), \tag{13}$$

where κ is a vector of bounded, continuous basis functions and $\hat{a}_i(t)$ is a parameter vector that is tuned according to an adaptation law. The existence of $\hat{\rho}_i(q,t)$ is guaranteed since any function over a bounded domain can be approximated arbitrarily well by some set of basis functions [Sanner *et al.*, 1992]. Next, for the control law

$$u_i = -k_p \frac{\partial J}{\partial p_i} \eta^{il}(p_i), \tag{14}$$

where k_p is a positive constant, $i \in \{1, \cdots, N\}$ and η^{il} is the i-th row, l-th column element of the inverse Riemannian metric matrix. We only need to revise the parameter k_p to a uniformly positive definite control gain matrix which may have a skew-symmetric component [Schwager *et al.*, 2008]. Then, the basic convergence of the control system is guaranteed by Theorem 1 in [Schwager *et al.*, 2007].

Since the environment is time-varying, we need to ensure the existence of the time-varying conformal mapping. Based on Corollary 1 in [Xu *et al.*, 2020], there always exists the time-varying diffeomorphism if this diffeomorphism is constructed by the Riemannian Mapping Theorem [Petersen, 2006]. The Uniformization Lemma is the generalization of the Riemannian Mapping Theorem [Petersen, 2006]. And the conformal mapping τ mentioned in the original text is constructed by the Uniformization Lemma, which simplifies the study of simply connected Riemannian surfaces by reducing it to the study of the disk, plane, or sphere. Hence, the existence of the time-varying conformal mapping is guaranteed. Above all, the proposed framework can be adapted to the dynamic environments whose density varies slowly.

2 Simulation Results

2.1 Computational cost

To estimate the computational cost, we analyze the time complexity of the designed algorithms and record the time for each iteration in the simulation. At first, we set the parameters as follows: The agent number is N, the number of the whole vertices is v, and the number of the vertices corresponding to the i-th agent is v_i. The number of iterations of the Iterative Closest Point(ICP) is k. For per agent, the time complexity of Delaunay triangulation is $O(v_i log v_i)$. And the harmonic mapping H_i can be easily discretized using the cotangent formulation, whose time complexity is $O(v_i)$ [Choi, 2021]. We use the Linear Beltrami Solver(LBS) method to construct the rectangular mapping ϑ_i^* and ϑ^*. The LBS method essentially solves a linear system, so the time complexity of the LBS method is $O(v_i)$ [Lui *et al.*, 2013].

The time complexity of the exponential mapping ι is $O(v_i)$. And we use the LBS method to construct the quasi-conformal mapping ϖ, whose time complexity is $O(v_i)$. So the time complexity of the composition mapping $\varpi \circ \iota \circ \vartheta^*$ is $O(v_i)$.

Then, we analyze the time complexity of the designed Mapping Construction Algorithm. The time complexity of the parallel computing of agents (step 1-10) is $O(Nvlogv)$. The time complexity of choosing μ_*(step 11)is $O(N)$. The time complexity of the LBS method(step 12) is $O(v)$. The time complexity of the inverse rectangular mapping of the multi-agents(steps 13-17) is $O(Nv)$. The time complexity of ICP(step 18) is $O(kvlogv)$ [Besl *et al.*, 1992]. The time complexity of the global Delaunay triangulation(step 19) is $O(vlogv)$. The time complexity of the composition mapping $\varpi \circ \iota \circ \vartheta^*$ is $O(v)$. Hence, the time complexity of the designed Mapping Construction Algorithm is

$$O(Nv\log v + kv\log v + Nv) = O((N+k)v\log v) \tag{15}$$

Finally, we analyze the time complexity of the designed Coverage Optimization Algorithm. The time complexity of running Algorithm 1(step 1) is $O((N+k)v\log v)$. For the while loop(steps 4-15), we update ψ_i, ψ_i', p_i and p_i' when $t < T_\varepsilon$. The time complexity of the whole loop is $T_\varepsilon \cdot O(1)$. The time complexity of the multi-agent communication integration(steps 19-23) is $O(N)$, and the computation of the optimization performance J_i^k(step 24) is $O(N)$. Hence, the time complexity of the main for loop(steps 2-25) is $K^* \cdot (T_\varepsilon \cdot O(1) + O(N) + O(N))$. The time complexity of the selection of the optimal solution k^* is $O(K^*)$. Thus, the complexity of the Coverage Optimization Algorithm is

$$O((N+k)v\log v) + K^*(T_\varepsilon \cdot O(1) + O(N)) + O(K^*) = O((N+k)v\log v + K^*(T_\varepsilon + N)) \tag{16}$$

In the simulation, we initialize the basic parameters as follows: $K^* = 30$, $N = 4$, $T_\varepsilon = 10$, $k_\psi = 0.03$, $k_p = 0.1$, $v = 7344$, and $k = 25$. And we define the convergence indexes as follows: n is the number of convergence iterations, t is the convergence time, $RMSE$ is the Root Mean Square Error and $Inbalance\ degree$ is the imbalance of the workload, which is calculated using $Inbalance\ degree = \max(m)\backslash\min(m)$($m$ is the workload array). It's noteworthy that the convergence time t is the total convergence time of the multi-agent system. Simulation results are presented in Fig. 1. In particular, Fig. 1 illustrates the number of convergence iterations n is 30, the total convergence time t is 14.4705 s, $RMSE = 3.3696e-11$ and $Inbalance\ degree = 1$.It means that the designed Coverage Optimization Algorithm can be efficiently computed and the approach is feasible for real-time operation in practice.

2.2 Quantitative metrics

First, we add multi-agent trajectories and density distributions to the initial simulation image. And the new simulation results are showed in Fig. 2. The results demonstrate the coverage control approach can

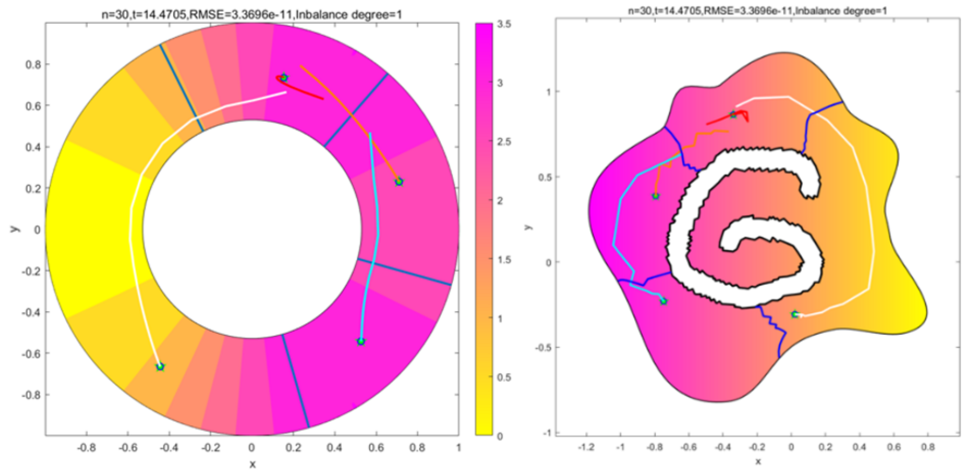

Figure 1: The convergence indexes of the coverage optimization algorithm.

accomplish the coverage mission and achieve workload balance in the original region (Ω-O)$\cup\partial O$ and the mapped region Ξ. And we record the convergence iterations n, convergence time t, $RMSE$ and the $Inbalance\ degree$ in Fig. 3. It's noteworthy that the convergence time t is the total convergence time of the multi-agent system. The simulation results illustrate the number of convergence iterations n is 28, the total convergence time t is 13.5802 s, $RMSE = 2.3177e-15$ and the $Inbalance\ degree = 1$. Then, we calculate the control strength u, workload m and phase angle ψ of the multi-agents. Fig. 4 illustrates that the control strength u of the multi-agents will gradually become 0 as time goes by. It means that the designed control law drives multi-agents to the corresponding optimal centroid. (p.s. Fig. 4 exhibits a sudden change in phase angle. The reason for this is that the partition bar rotated and once again crossed the 0-degree reference line.) Additionally, the workload of the multi-agents will gradually converge to 1.1944, which illustrates the effectiveness of the designed partition algorithm. We have recorded the varieties of the phase angles over time. The simulation results show the convergence of the designed partition algorithm.

2.3 Comparisons with baseline approaches

Next, we compare the designed algorithm with the circular search algorithm designed in [Zhai *et al.*, 2023]. We set the density of the obstacle O to zero. Then, we directly use the circular search algorithm to partition the region (Ω-O)$\cup\partial O$. The simulation result is shown in Fig. 5. It illustrates that the partition par collides with the edge of the obstacle, and it forms the disconnected sub-regions. But the designed algorithm achieves the coverage task and workload balance, which illustrates the effectiveness of the

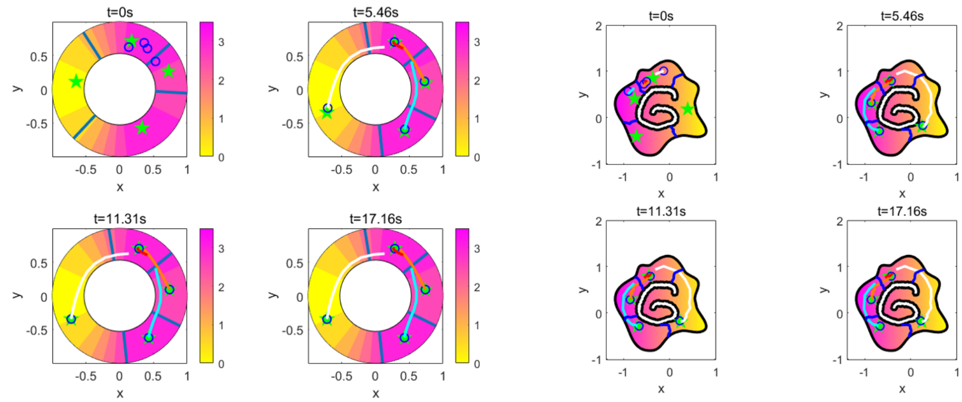

Figure 2: Coverage process on the mapped region Ξ and the original region $(\Omega\text{-}O)\cup\partial O$.

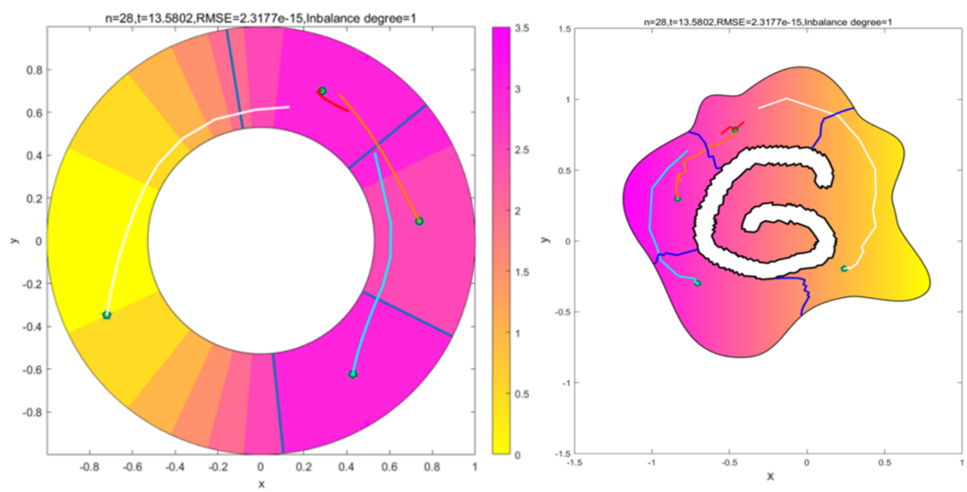

Figure 3: The convergence indexes of the coverage process.

designed algorithm.

2.4 Multiple obstacles

Then, we provide an explicit extension of our methodology to handle multiple obstacles. Firstly, we analyze the mapping design process. Analogous to the Koebe's iteration method [Koebe *et al.*, 1910], the method handles the k holes of the surface Ω one by one. For the i-th hole($i = 1, \cdots, k$), fill all but the i-th hole. Then, we solve for an annulus map f_i using the mapping construction algorithm with the quasi-conformal composition step ϖ skipped, which is used for the one-hole environment. Finally, we remove all the filled holes. Since these steps may not follow the distribution of the holes on Ω well, the area distortion of the map may be large. To solve this problem, we adopt the Mobius area

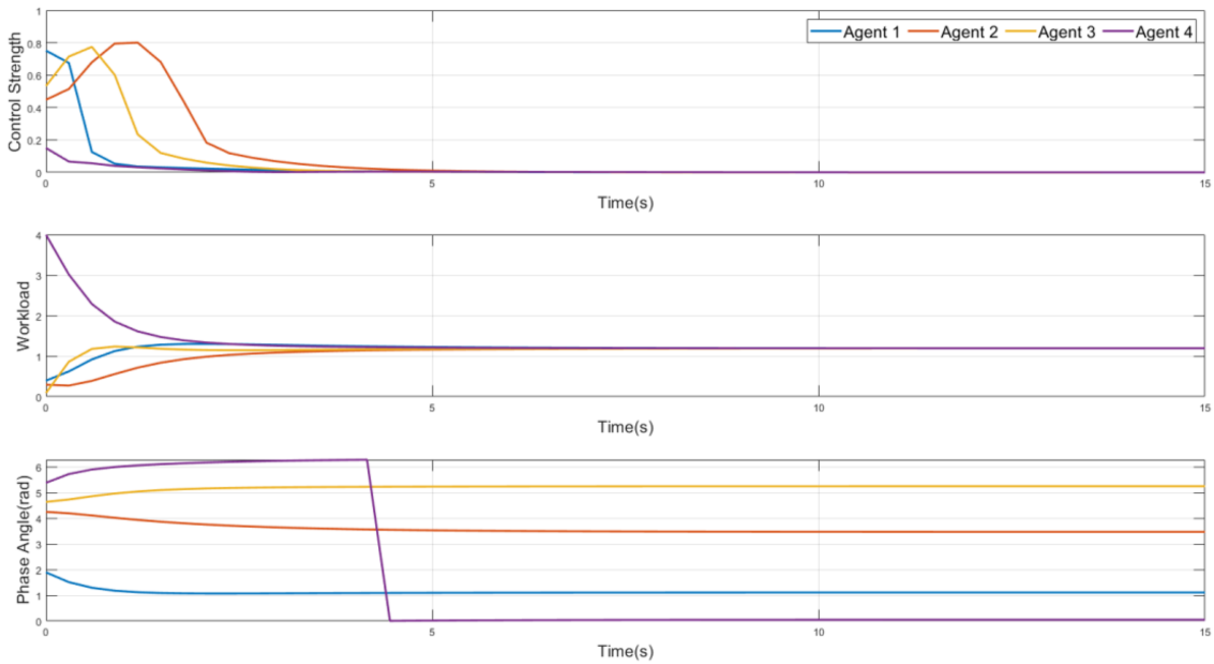

Figure 4: The convergence indexes of the coverage process.

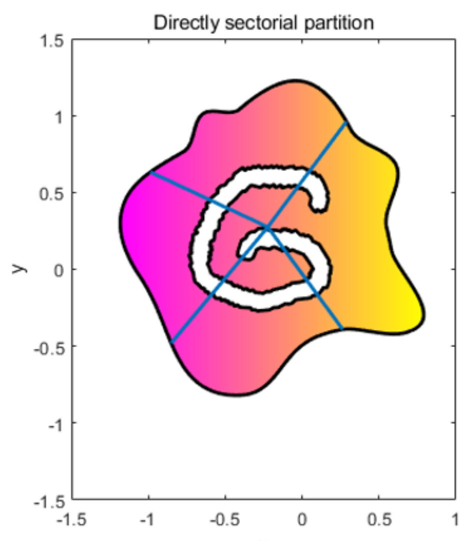

Figure 5: The partition result of the circular search algorithm in [Zhai *et al.*, 2023].

correction scheme [Mobius *et al.*, 2020] to reduce the area distortion. All the k holes become close to circles under the above steps, but they may not all be perfectly circular. Thus, for each hole, we find the maximum inscribed circle of it and project all boundary vertices onto this circle [Choi, 2021]. After this operation, we obtain a unit disk with k circular holes. Additionally, we use the LBS method to obtain

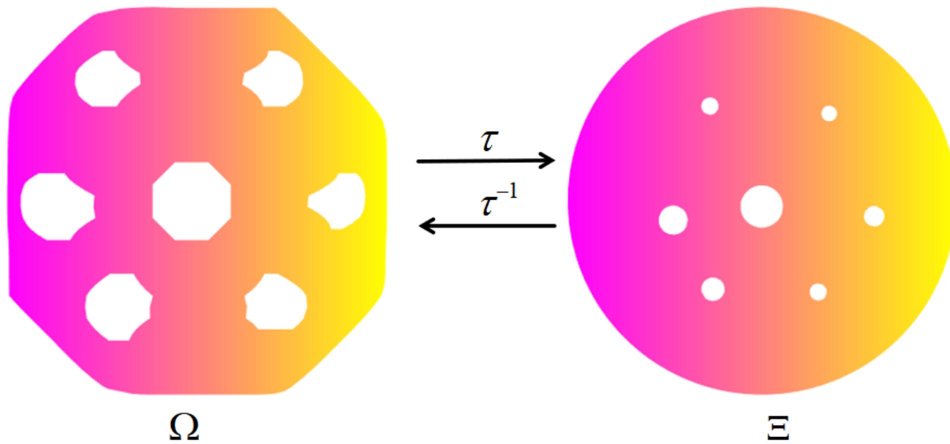

Figure 6: The conformal mapping which is suitable for the multiple obstacles environment.

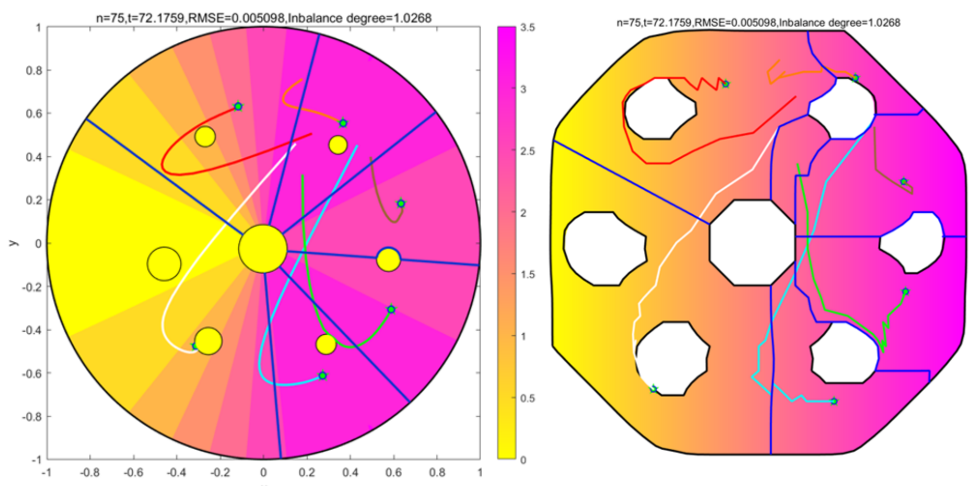

Figure 7: The coverage process and its indexes in the multi-obstacle region.

the quai-conformal mapping ϖ, which is used to reduce the angle distortion [Lui *et al.*, 2013]. Above all, we obtain the conformal mapping τ, which is suitable for the multiple obstacles environment. The conformal mapping result is shown in Fig. 6. The conformal mapping for multiple obstacles involves two rotations, so the density distribution of the original and mapped regions is the same, which is different from the one-obstacle situation.

Next, we focus on the design of the partition algorithm. Suppose the number of agents is N. To balance the workload among sub-regions, the dynamics of the partition phase remain designed as

$$\dot{\psi}_i = k_{\psi}\left(m_i - m_{i-1}\right), \tag{17}$$

where k_{ψ} is a positive constant and $i = 1, \cdots, N$. We set the density of the obstacles to 0. Thus, the

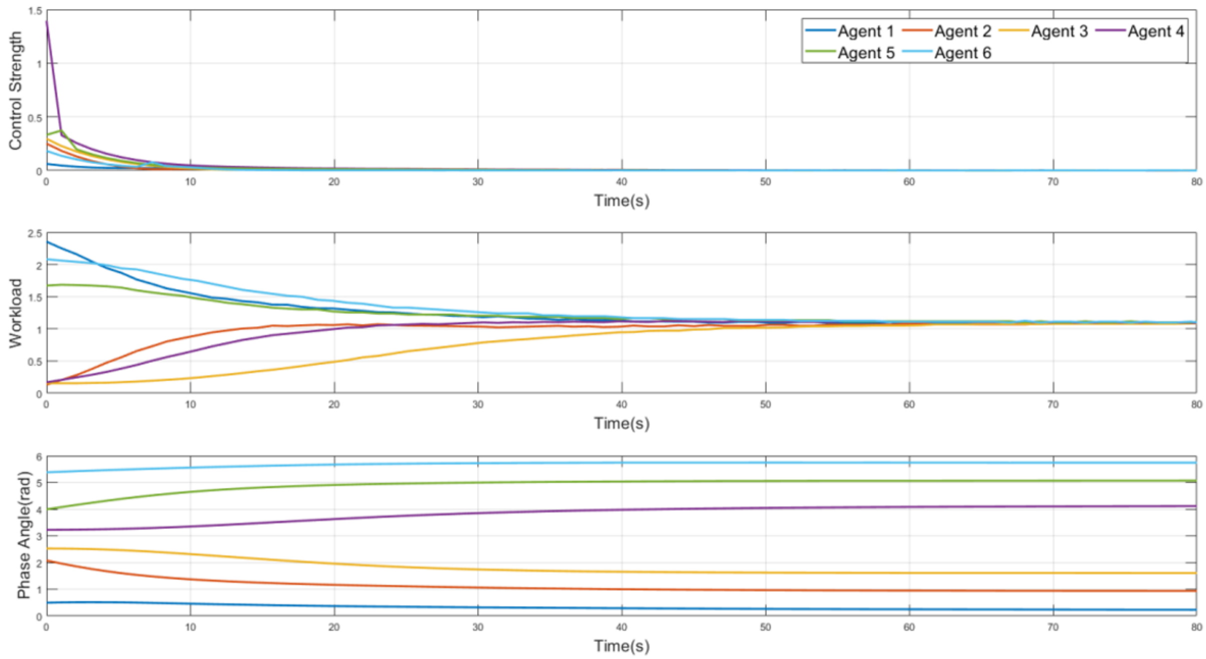

Figure 8: The coverage process and its indexes in the multiple obstacles environment.

color of the obstacles in the mapped region is yellow. Now we need to solve two problems: The partition bars will pass through one or more obstacles. (i.e., Problem 1). If two or more partition bars pass through the same obstacle simultaneously, disconnected sub-regions will occur. (i.e., Problem 2). For Problem 1, since the obstacles are all circular regions and the coordinates of the obstacle boundary can be determined by the mesh boundaries algorithm [Choi, 2021], it's easy to obtain the center and radius of each obstacle. Hence, it becomes easier to calculate the boundary arc of the obstacles. Then, we can design the partition bar for the obstacle-crossing section to follow the boundary of the obstacle, which won't change the workload of the corresponding sub-region. For Problem 2, based on the solution to Problem 1, the center and radius of each obstacle are easy to obtain. We can obtain the concentric circles of each obstacle. When multiple partition bars pass through the same obstacle, in order to avoid forming disconnected sub-regions, we require the partition bars to make detours on concentric circles that are slightly larger than the obstacle itself. (The partition bars rotate counterclockwise, and the ones that pass first make detours on the larger concentric circles.) Additionally, the size of the concentric circles needs to be strictly determined so as to minimize the impact on workload. The details of this technique will be illustrated in my future work.

Then, we analyze the design of the control law. We can adopt the control law and the dynamics of multi-agent systems as presented in the original text. However, due to the change in the topological structure of the environment, the calculation of the Cut locus C_{p_i} and the Riemannian metric η of this area will also change. The specific calculations and definitions will be explained in my future work. The control input u_i have the same form as

$$u_i = -k_p \frac{\partial J}{\partial p_i} \eta^{il}(p_i), \tag{18}$$

where k_p is a positive constant, $i \in \{1, \cdots, N\}$ and η^{il} is the i-th row, l-th column element of the inverse Riemannian metric matrix. The dynamic of the i-th agent is presented as

$$\dot{p}_i = u_i. \tag{19}$$

The simulation results illustrate the number of convergence iterations n is 75, the total convergence time t is 72.1759 s, $RMSE = 0.005098$, and $Inbalance\ degree = 1.0268$. Fig. 7 demonstrates that the agent can arrive at the corresponding optimal centroid while avoiding the multiple obstacles. In the meantime, the algorithm achieves the workload balance. Benefit from the solution to Problem 2, disconnected sub-regions will not be formed.

Finally, for the larger environment, the area of the region hasn't affected the partition cost. It only increases the distance between the agent position and the optimal centroid. So our focus of analysis was placed on the multiple obstacles environment. For the multiple obstacles environment, we still use the coverage performance indexes in the single-hole scenario (e.g., the convergence iterations n, convergence time t, $RMSE$, $Imbalance$, control strength, the change of workload, and phase angle) to demonstrate the effectiveness of the coverage optimization algorithm, which is suitable for the multiple obstacles environment. The simulation results are in Fig. 7 and Fig. 8. Fig. 7 demonstrates that the agent can arrive at the corresponding optimal centroid while avoiding the multiple obstacles. In the meantime, the algorithm achieves the workload balance. Benefit from the solution to problem 2, disconnected sub-regions will not be formed. The simulation results illustrate that the number of convergence iterations n is 75, and the total convergence time t is 72.1759 s, $RMSE = 0.005098$, and $Inbalance\ degree = 1.0268$. The control strength u of the multi-agents will gradually become 0 as time goes by. The workloads and the phase angles will gradually converge to a constant. We compared these performance indexes with the single-obstacle environment involving 8 agents(n=66, t=48.3997s, $RMSE$=4.616e-05, and $Inbalance\ degree = 1.0005$). When there are fewer agents, the number of convergence iterations n, $RMSE$, $Imbalance$ are bigger, and the total convergence time t is longer. It demonstrates the complexity of the multiple obstacles environment and the real-time nature of the control algorithm.

Additionally, compare our method with the Hierarchical Coverage Control(HCC) algorithm mentioned in [Zou *et al.*, 2025]. The HCC algorithm applies only to environments where multiple obstacles

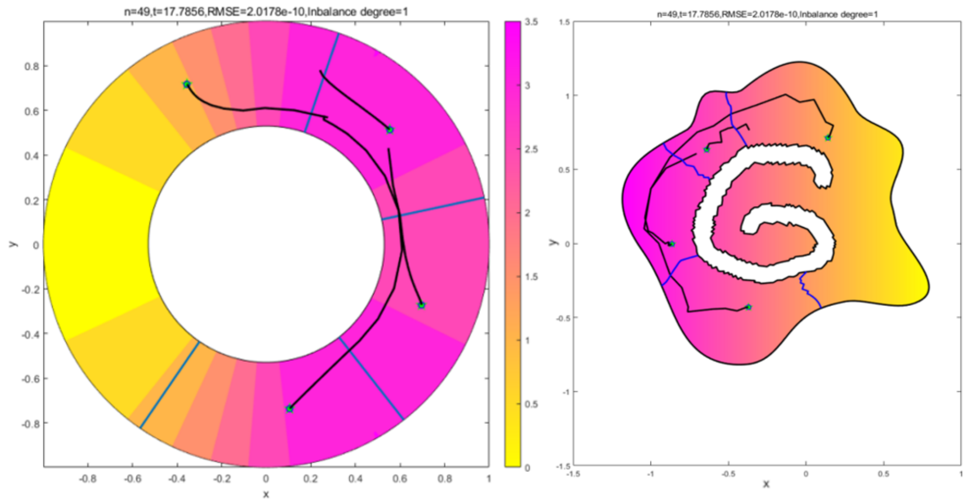

Figure 9: The convergence indexes of the coverage process involving 4 agents.

can be placed in a single form within the Voronoi cells. In the meantime, the dynamics model with non-holonomic constraints of the MAS can't achieve collision avoidance with these obstacles. However, our algorithm is applicable to multiple obstacles of any shape in a two-dimensional plane, and it can also enable multi-agent obstacle avoidance. Then, we compare our work with the latest research on the multiple obstacles avoidance with a huge number of agents [Surendhar *et al.*, 2025]. The Obstacle Avoidance Coverage Control Algorithm mentioned in this thesis is only suitable for the convex region, which can be divided by the Voronoi partition. And the obstacles can only be circular. However, our method could handle multi-obstacle areas of any shape, and there are no requirements for the specific area itself.

2.5 Cost scaling with the number of agents.

We provide the discussion and simulations about the different numbers of agents and obstacle numbers. For simplicity, we have unified all the paths of the multi-agents into black. We remain to adopt the convergence iterations n, convergence time t, $RMSE$, and the $Inbalance\ degree$ to represent the computational cost, and record the control strength u, workload m, and phase angle ψ of the corresponding situation. It's noteworthy that the convergence time t is the total convergence time of the multi-agent system. For the designed coverage optimization algorithm, we set the number of multi-agents to 4, 8, and 16, respectively. And we recorded the corresponding coverage performance indexes. For 4 agents, the simulation results are shown in Fig. 9 and Fig. 10. For 8 agents, the simulation results are shown in Fig. 11 and Fig. 12. For 16 agents, the simulation results are shown in Fig. 13 and Fig. 14. Then, we record the coverage performance indexes in Table 1.

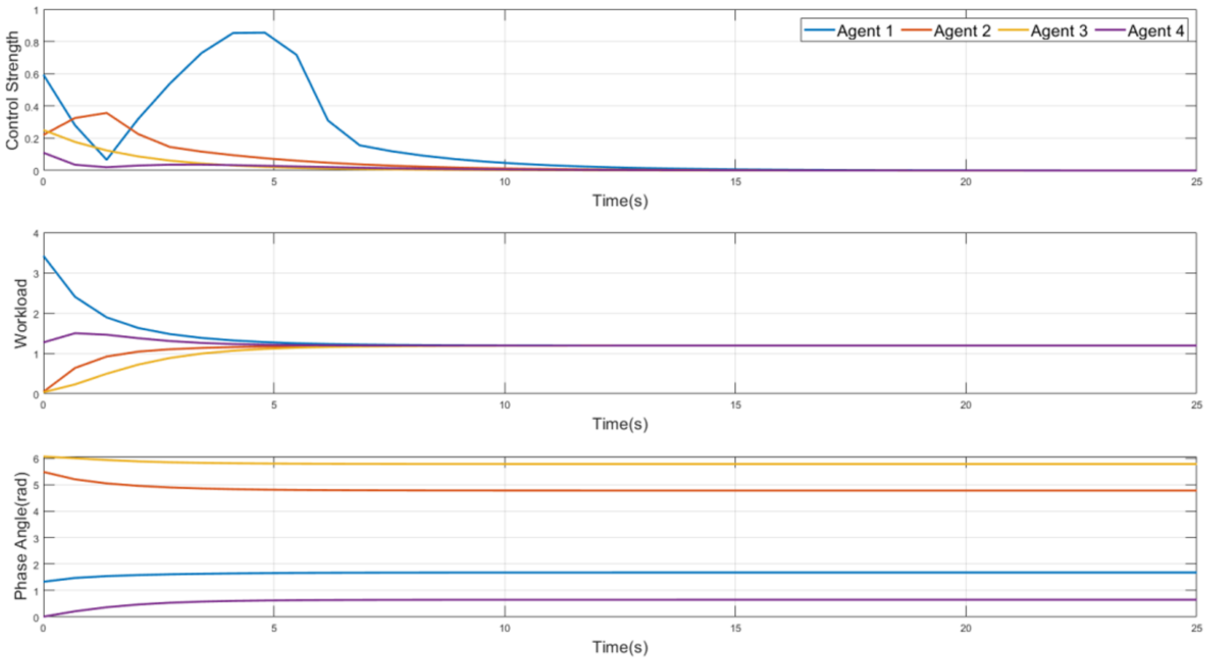

Figure 10: The convergence indexes of the coverage process involving 4 agents.

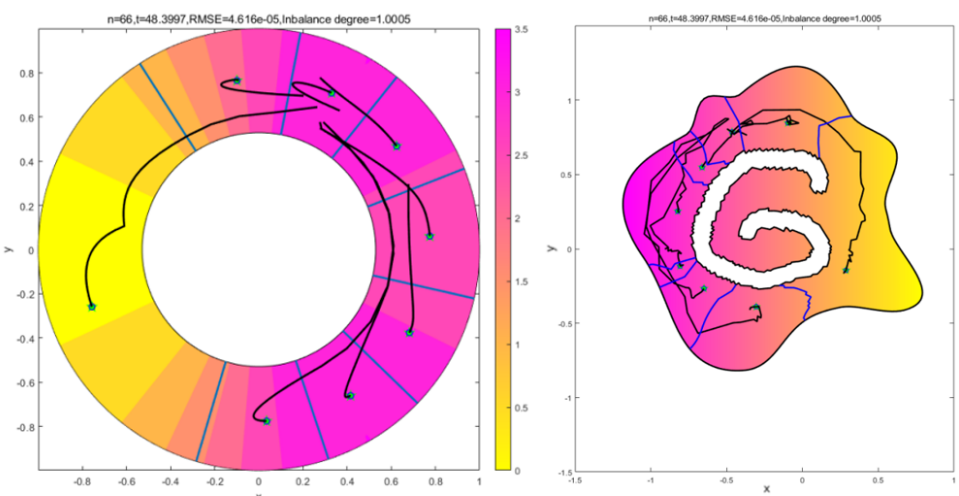

Figure 11: The convergence indexes of the coverage process involving 8 agents.

As shown in Table 1, with an increasing number of agents, the convergence iterations n and convergence time t will increase. Additionally, $RMSE$ is still close to 0, and the $Imbalance$ is still close to 1. For the cases where the number of agents is 4, 8, and 16, the running time per agent per iteration is 0.0907s, 0.0917s, and 0.0795s, respectively. It illustrates that the computational cost of a single agent does not show significant changes as the number of agents increases, while still meeting the computational cost

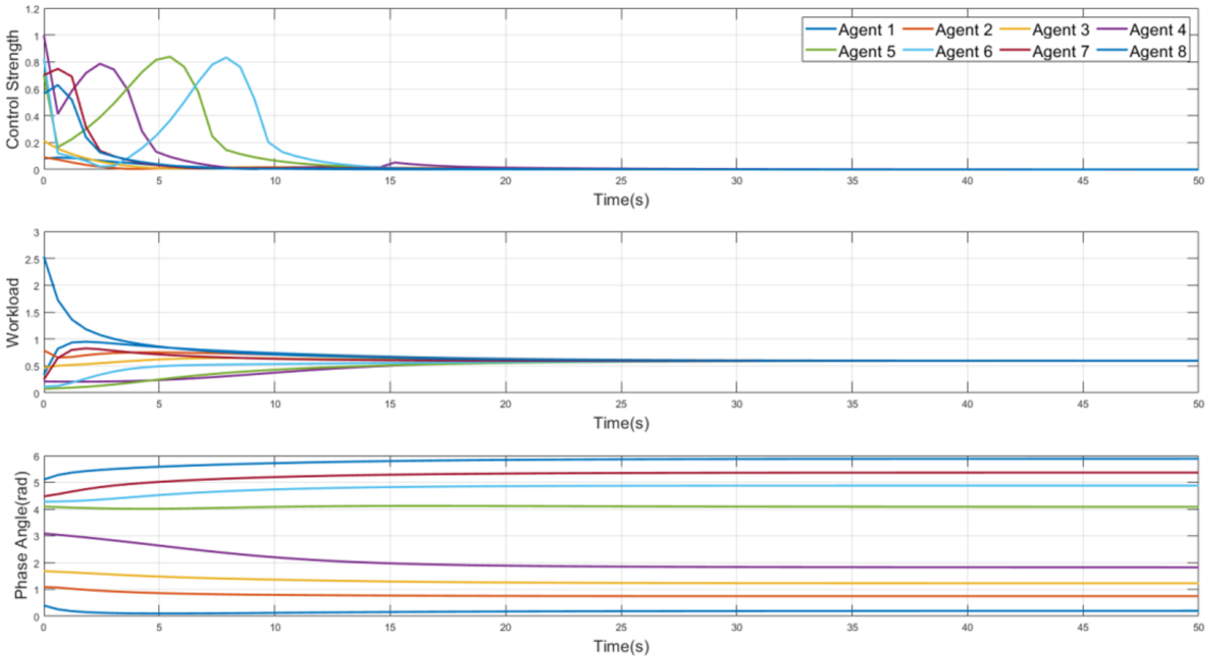

Figure 12: The convergence indexes of the coverage process involving 8 agents.

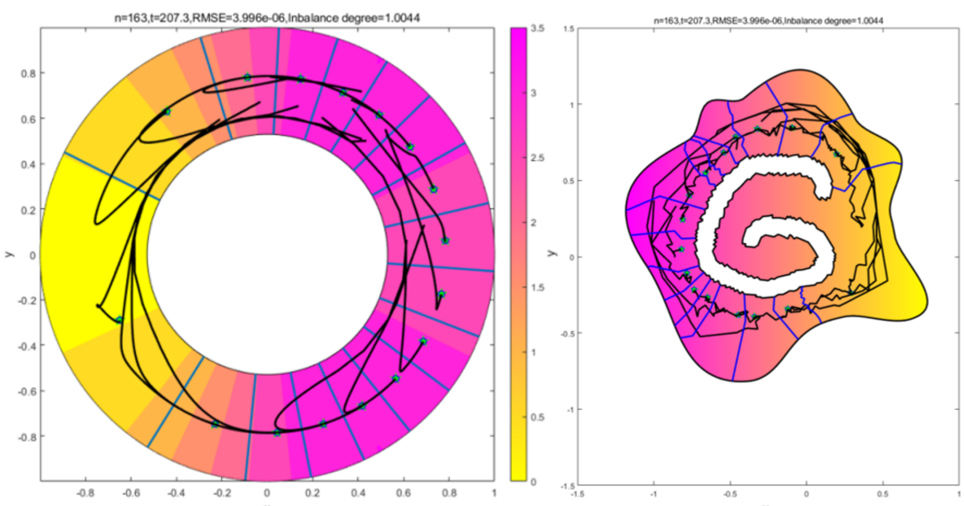

Figure 13: The convergence indexes of the coverage process involving 16 agents.

requirements. Then, Figure 10, Figure 12, and Figure 14 illustrate that the control strength u of the multi-agents will gradually become 0 as time goes by. Additionally, the workload of the multi-agents will gradually converge to a constant, and we have recorded the varieties of the phase angles over time. The results illustrate the effectiveness of the designed partition algorithm. Thus, the overall performance still meets the real-time requirement.

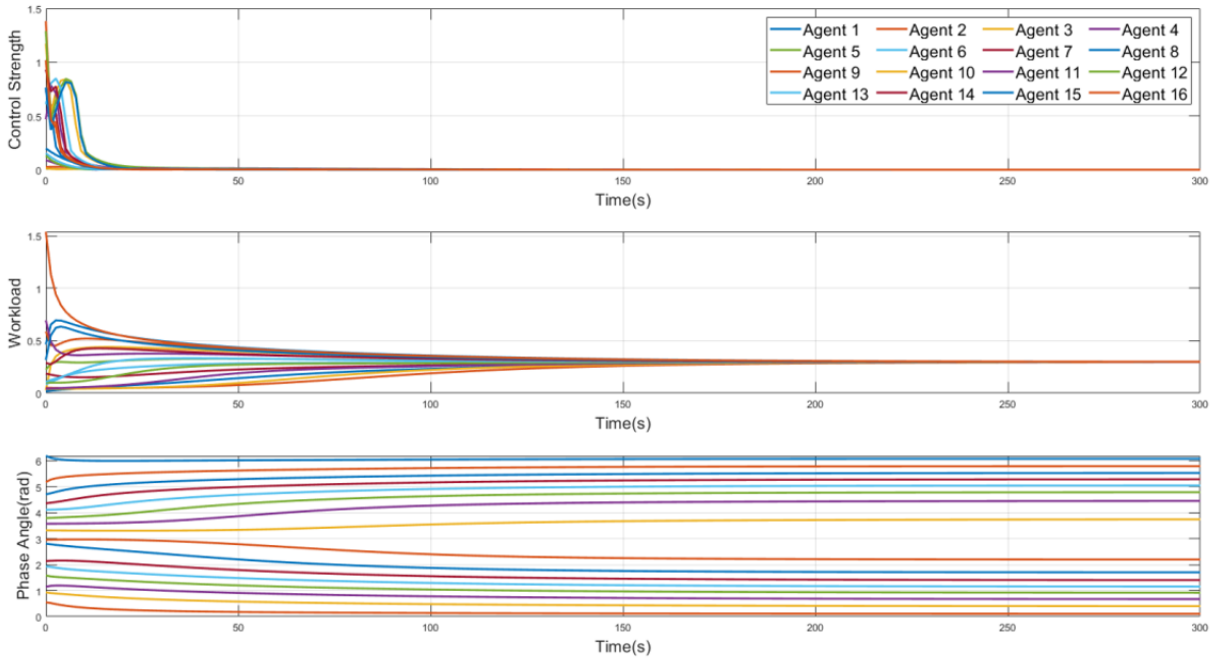

Figure 14: The convergence indexes of the coverage process involving 16 agents.

Table 1: The method changes with an increasing number of agents

Agent Number	Iterations	Convergence time	RMSE	Imbalance
4	49	17.7856s	2.017e-10	1
8	66	48.3997s	4.616e-05	1.0005
16	163	207.3000s	3.996e-06	1.0044

2.6 Agent failure

When the i-th agent fails, the $(i+1)$-th and $(i-1)$-th agents automatically become the new neighbors. The subregion originally corresponding to the i-th agent will re-participate in the distribution of quality, and the remaining agents will complete the tasks of coverage and workload balancing. The communication mechanism designed in the Coverage Optimization Algorithm(lines 18-23) guarantees the stability of the multi-agent system. The simulation results are shown in Fig. 15. When $t = 0$, all agents are cooperative in the workspace. And we suppose that the i-th agent fails at 0.1 seconds. Then, the remaining agents are still able to achieve coverage task and workload balance (see Fig. 15). It illustrates that the remaining agents are still capable of balancing workload and moving to the corresponding optimal centroid when an agent fails.

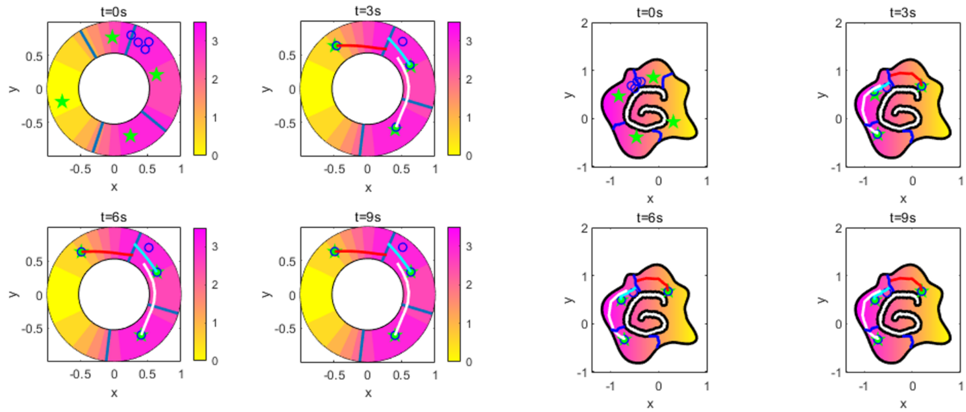

Figure 15: The coverage process in the case of an agent failing.

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